International Journal of Educational Research 133 (2025) 102666

Contents lists available at ScienceDirect

International Journal of Educational Research

journal homepage: www.elsevier.com/locate/ijedures

Pathways to equity: A mediation analysis of gender, SES, and mathematics achievement using PISA 2022 UK data

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ARTICLE INFO

Keywords:
Gender and mathematics achievement
Socioeconomic status
Growth mindset
Mathematics teacher support
Mathematics anxiety
UK mathematics education

ABSTRACT

This study examines the relationship between gender, socioeconomic status, and mathematics achievement in mathematics education in the United Kingdom, focusing on the mediating role of mathematics teacher support, growth mindset, and mathematics anxiety. Based on the data from the PISA 2022 database for the UK, the analysis uses the mediation effect test method. The results show no significant gender differences in mathematics achievement, while socioeconomic status has a positive effect on mathematics achievement. Male students exhibited receiving significantly higher levels of teacher support and demonstrating stronger growth mindset, but experiencing lower mathematics anxiety compared to female students. There is no correlation between socioeconomic status and mathematics teacher support, a positive correlation with growth mindset, and a negative correlation with mathematics anxiety. Research shows that teacher support and growth mindset positively affect mathematics achievement, while mathematics anxiety has negative effects. The study also finds that gender has a mediating influence on the path from gender to mathematics teacher support to growth mindset to mathematics anxiety and then to mathematics achievement. The study provides policy recommendations for improving the quality of mathematics education, especially for girls and students of low socioeconomic status, and emphasises the importance of enhancing teacher support, fostering a growth mindset, and reducing mathematics anxiety.

1. Introduction

Mathematics education in the United Kingdom has faced ongoing challenges. Results from the Programme for International Student Assessment (PISA) 2022 reveal a decline in students' performance in mathematics. Only 11 % of UK students achieve high scores in the subject, highlighting the need to investigate the underlying factors contributing to this issue (OECD, 2023).

One such factor may be students' mathematics learning anxiety. Previous studies have shown that anxiety significantly affects students' ability to succeed in mathematics (Pellizzoni et al., 2022). This anxiety may hinder students' engagement with mathematical concepts, reducing learning motivation and, in turn, leading to poorer performance (Živković et al., 2023). Prior research has indicated that various of factors, including gender and socioeconomic status (SES), influence students' level of mathematics anxiety (Svraka & Ádám, 2024). For instance, females often experience higher levels of mathematics anxiety compared to males, which negatively

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https://doi.org/10.1016/j.ijer.2025.102666
Received 6 April 2025; Received in revised form 18 May 2025; Accepted 4 June 2025
Available online 9 June 2025
0883-0355/

impacts their mathematics performance (Luttenberger, Wimmer, & Paechter, 2018). Similarly, students from higher SES background tend to have less mathematics anxiety and achieve better attainment in mathematics (Kalaycioglu, 2015).

Furthermore, growth mindset, in which students believe their abilities can improve with effort, is positively related to both reduced mathematics anxiety and improved achievement (Samuel & Warner, 2021). This suggests that fostering a growth mindset may help students overcome their learning anxiety and perform better in mathematics. Additionally, the support provided by mathematics teachers, whose roles (e.g., teaching, guidance, and encouragement) are crucial in mathematics classes, may shape students' growth mindset and mathematics anxiety (Sherwin, 2020). However scarce research has been conducted, particularly within the UK context.

The current study utilises the PISA 2022 data to investigate how gender, SES, teacher support, growth mindset, and mathematics anxiety interact to affect mathematics achievement in the UK. A mediating effect analysis will be conducted to address three main aims: first, to explore the direct effects of gender and socioeconomic status on math achievement, and investigate gender and SES differences in mathematics teacher support, growth mindset and mathematics anxiety; second, to examine the influences of mathematics teacher support, growth mindset, and mathematics anxiety on mathematics achievement; and third, to investigate how mathematics teacher support influences achievement through a chain mediation effect from gender and SES, involving growth mindset and mathematics anxiety. The findings will be useful for policymakers in designing specific strategies to address gender and socioeconomic disparities in mathematics learning. Additionally, the emphasis on teacher support will help school administrators and mathematics teachers on how to enhance teacher-student interactions to reduce mathematics anxiety, improve students' growth mindset, and foster a positive learning environment.

2. Literature review

2.1. Theoretical foundations

2.1.1. Growth mindset: from belief to achievement

The growth mindset is a central concept in Dweck's Mindset Theory (2006). It suggests that those who view their intelligence and abilities as malleable (growth mindset) are more likely to embrace challenges than those who perceive traits as fixed (fixed mindset). That is, these individuals can persevere through setbacks and achieve higher levels of academic success. For instance, a student with a growth mindset interprets mathematical challenges as opportunities for improvement, fostering persistence and reducing fear of failure (Yeager & Dweck, 2020).

Empirical studies in mathematics education confirm that growth mindset interventions improve achievement indirectly through mechanisms like self-efficacy and failure attribution. For example, a study demonstrated that students with a growth mindset attribute poor performance to effort rather than fixed ability, which enhances their confidence to tackle difficult problems (Dong et al., 2023). Meta-analytic evidence further supports this, showing that growth mindset interventions yield stronger effects in mathematics than in other subjects, particularly when combined with teacher support (Burnette et al., 2023).

2.1.2. Control-value theory: emotions as mediators of achievement

Control-Value Theory (Pekrun, 2006) provides a complementary framework to explain how students' emotions, particularly anxiety, influence academic achievement. The theory identifies two key dimensions: control appraisals and value appraisals. Control appraisals describe the impact of students' beliefs about their abilities on learning outcomes (e.g., Can I solve this problem with effort?). Value appraisals refer to students; perceptions of the importance or utility of a task (e.g., Is math worth investing time in?).

High control and positive value appraisals reduce anxiety and increase engagement, whereas low control or negative value appraisals cause anxiety and avoidance behaviors. For instance, students who perceive math as uncontrollable or irrelevant are more likely to experience crippling anxiety (Pekrun, 2006). Control-Value Theory was applied to mathematics education and showed that if teachers matched task difficulty to student ability (enhanced control) and emphasised real-world applications (enhanced value), they significantly reduced anxiety (Klee et al., 2022).

2.1.3. Growth mindset and control-value theory: a synergistic framework

The interplay between growth mindset (Dweck, 2015) and Control-Value Theory (Pekrun, 2006) offers a powerful perspective on how students' beliefs and emotions affect mathematics achievement. Growth mindset – the belief that one's abilities can be developed through effort (framing effort as a tool for improvement) – enhances students' control appraisal dimension of Control-Value Theory. Meantime, fostering value appraisal – such as highlighting the real-world applications of mathematics – can motivate students to adopt a growth mindset, as they perceive effort as meaningful rather than wasteful. For example, teachers who emphasise mastery over grades (value appraisal) and provide actionable feedback (control of appraisal) create environments that allow growth mindsets to thrive (Chen et al., 2024).

2.2. Students' individual characteristics and their learning outcomes

2.2.1. Socioeconomic status

Socioeconomic status (SES) is defined as a metric that assesses an individual's combined economic and social standing, with three commonly recognized indicators: education, income, and occupation (Baker, 2014). Students' SES refers to the educational background, income level, and occupational status of their primary family members, particularly their parents (Arztmann et al., 2024). Existing research consistently indicates that the overall trend in the relationship between SES and mathematics achievement demonstrates a significant positive correlation (İdil et al., 2024). SES influences students' mathematics achievement through differences in access to educational resources, quality of schooling, family support, and psychological factors, which collectively shape learning opportunities and outcomes (Wang et al., 2014).

2.2.2. Gender

Studies in the past have generally found that boys do better than girls on mathematics tests, especially in high school (Tsui, 2007). More specifically, girls have shown a slight advantage in computational performance during primary and middle school, while boys have demonstrated a more pronounced superiority in mathematics during high school and university (Hyde et al., 1990). As society evolves, researchers have observed a gradual narrowing of gender disparities in mathematics achievement, driven by the changing of societal expectations, self-perception, and access to educational support and opportunities (Lindberg et al., 2010). A recent study involving 13 million students indicates that there is already no significant gender gap in mathematical achievement in England, though male students demonstrate more polarized results, more frequent among both the highest and lowest achievers (Coates, 2025).

2.3. The impact of SES and gender on teacher support

Mathematics teacher support refers to the assistance and guidance provided by teachers in emotional, cognitive, and autonomy-related aspects, aimed at enhancing students' motivation, understanding, and academic achievement in mathematics (Fisher & Royster, 2016). Research has demonstrated that mathematics teacher support significantly promotes students' academic achievement by fostering a positive learning environment, improving conceptual understanding, and boosting self-efficacy in mathematics (Chang, 2015). However, it is important that the effects of mathematics teacher support may vary based on students' gender and SES. Teachers, influenced by gender stereotypes that associate mathematics ability more strongly with males, may unconsciously provide more mathematics support and encouragement to male students than to female students (Dersch et al., 2022). Students from low SES often enter school with limited access to early mathematics resources, resulting in weaker foundational skills, which may lead teachers to disproportionately allocate support and attention to students from higher SES who are already better prepared (Cesnaviciene et al., 2022; LeGere, 2023). However with the growing emphasis on educational equity and the integration of fairness training in teacher preparation programs, the disparities in mathematics teacher support based on gender and SES are gradually diminishing (Liao et al., 2022). The specific nature of these disparities in the UK remains underexplored and requires further investigation.

2.4. The mediating role of teacher support in the student learning process

Research on the role of mathematics teacher support in mediating gender and SES effects on growth mindset and anxiety is limited, but existing studies provide valuable insights. Teacher gender stereotypes often lead to more effective support for boys, such as tailored encouragement and problem-solving guidance (Gunderson et al., 2018), which can alleviate mathematics anxiety, though this effect is more pronounced for boys (Wang et al., 2024). Similarly, higher SES students tend to receive greater mathematics teacher support, such as personalized feedback and resources, compared to their lower SES peers (Bakchich, 2024). This support positively influences growth mindset by fostering students' belief in their ability to improve through effort (Vestad & Bru, 2024), which in turn reduces mathematics anxiety as students view challenges as opportunities rather than threats (Dong et al., 2023). Additionally, mathematics teacher support indirectly affects mathematics anxiety (Li et al., 2021). However, the pathways linking (a) gender to mathematics teacher support, growth mindset, and anxiety, and (b) SES to mathematics teacher support, growth mindset, and anxiety remain underexplored. This study addresses these gaps by examining how mathematics teacher support mediates the effects of gender and SES on growth mindset and anxiety, offering new insights into the dynamics of mathematics learning and the potential for equitable teacher interventions to address disparities.

2.5. Contextualising the framework in UK mathematics education

In the UK, mathematics education has long been a focal point of policy and research, particularly concerning equity and achievement gaps. The intersection of student gender and SES with mathematics achievement has been widely documented, with boys often outperforming girls in standardized assessments, and students from higher SES backgrounds consistently achieving higher scores than their lower SES peers (Department for Education, 2021). These disparities highlight the importance of examining individual student characteristics, such as gender and SES, as predictors of mathematics achievement.

Recent studies in the UK have also emphasized the role of student learning processes in shaping these outcomes. Teacher support, for instance, has been identified as a critical factor in mitigating the negative effects of low SES on mathematics performance (Yu & Singh, 2018). Similarly, growth mindset interventions have shown promise in addressing gender-related achievement gaps by fostering resilience and motivation among female students (Degol et al., 2018). However, the persistent issue of mathematics anxiety, particularly among girls and students from disadvantaged backgrounds, continues to undermine these efforts, suggesting a need for further exploration of its mediating role (Wang, 2020).

This framework aligns with the broader educational priorities in the UK, which aim to reduce inequality and improve outcomes for all students. By situating student gender and SES within individual characteristics, and examining their direct and indirect effects through teacher support, growth mindset, and mathematics anxiety, this study offers a comprehensive lens for understanding the complex dynamics of mathematics achievement in the UK context. Such an approach not only addresses existing gaps in the literature but also provides actionable insights for policymakers and educators striving to create more equitable learning environments.

2.6. The current study

In sum, the present research conceptualizes the relationship between gender, socioeconomic status, mathematics teacher support, growth mindset, mathematics anxiety, and mathematics achievement. Based on the Control-Value Theory, Growth Mindset theory, and previous studies, this paper hypothesizes that gender and socioeconomic status, as student individual characteristics can influence mathematics achievement, and that mathematics teacher support, growth mindset, and mathematics anxiety, as student learning process serve as mediators in this relationship. To be precise, the current research will focus on addressing the following key research questions:

RQ1: How do gender and socioeconomic status directly influence mathematics achievement? Are there gender and SES differences in teacher support, growth mindset, and anxiety?

RQ2: How does students' learning process impact math achievement, after controlling their characteristics?

(1) What are the relationships between teacher support, growth mindset, mathematics anxiety and achievement?
(2) To what extent does mathematics teacher support contribute to the mechanisms linking gender, socioeconomic status, and mathematics achievement, particularly through its influence on growth mindset and mathematics anxiety?

3. Method

3.1. Data

Data for this analysis were obtained from the 2022 Programme for International Student Assessment (PISA), administered by the Organisation for Economic Co-operation and Development (OECD). PISA is a large-scale international survey concentrating on the competencies of 15-year-olds in the subjects of mathematics, reading, and science. The main subject of PISA 2022 a particular research focus on mathematics. The sample included 12,972 students (6575 boys and 6397 girls) from 451 schools.

3.2. Variables

3.2.1. Student individual characteristics

In this study, student individual characteristics encompass gender and socioeconomic status.

Gender was considered as a variable, utilising "0″ for Female and "1" for Male within the PISA 2022 database before Mplus 8.5 (Muthén & Muthén, 2017) analysis.

Regarding socioeconomic status in PISA 2022, it was assessed using the Economic, Social, and Cultural Status (ESCS) index (OECD, 2023), derived from highest parental occupation status, highest education of parents in years, and home possessions (OECD, 2023, p. 407). To acquire the results, students completed comprehensive questionnaires in the Student Questionnaire covering these three dimensions.

The detailed item information of gender and ESCS (Table A1) is shown in Appendix A.

3.2.2. Student learning process

This study conceptualizes the student learning process as comprising three key components: mathematics teacher support, mathematics anxiety, and growth mindset.

PISA 2022 defines mathematics teacher support as in-class assistance from educators, excluding extracurricular activities, measured by scale ST270 (e.g., "How often: The teacher shows an interest in every student's learning"), which employs a 4-point Likert-Scale ranging from "1 = every lesson" to "4 = never or almost never".

Mathematics anxiety is assessed in PISA 2022 by evaluating students' agreement with statements on mathematical attitudes, represented by scale ST292 (e.g., "Agree/disagree: I often worry that it will be difficult for me in mathematics classes"), which utilizes 4-point Likert-Scale ranging from "1 = strongly agree" to "4 = strongly disagree".

PISA 2022 focuses on growth mindset, the belief that intelligence can improve through effort, measured by students' agreement with statements on attitudes towards intelligence and effort, using scale ST263 (e.g., "Agree/disagree: Your intelligence is something about you that you cannot change very much"), which applies 4-point Likert-Scale ranging from "1 = strongly disagree" to "4 = strongly agree".

3.2.3. Mathematics achievement

Plausible values are employed to assess mathematics achievement in PISA 2022. According to the PISA 2022 Technical Report (OECD, 2023), these values are derived from normal approximations to posterior distributions, estimated using multivariate latent regression models. Specifically, ten plausible values (PV1MATH to PV10MATH) were generated. The estimation process was conducted using Mplus 8.5 (Muthén & Muthén, 2017), with the TYPE=IMPUTATION command averaging across ten datasets to obtain accurate estimates and standard deviations.

All item information (Tables A1) and statistical descriptive analysis (Table A2) are shown in Appendix A.

3.3. Statistical analytical procedure

Mediational analyses are prevalent for addressing the fundamental research question. In this study, three variables meet David's (2024) criteria for multiple regression, consistent cases, and shared covariates across equations. Hence, they are suitable for mediation analysis.

Fig. 1 illustrates the direct impact of student individual characteristics (causal variable) on mathematics achievement (outcome), termed the total effect (path c).

Fig. 2 depicts the study's mediation analysis hypothesis model, with path a representing the influence of student individual characteristics on the student learning process and path b representing the impact of the student learning process on mathematics achievement. Together, paths a and b constitute indirect effects, while path c' denotes the direct effect. According to David (2024), paths c, a, b, and c' satisfy the equation $c = c' + ab$, and the research will be conducted based on this model.

Standardized coefficients for direct (Table A3) and standardized cofficients for indirect effects (Table A4) were derived from structural equation modeling. Bootstrap confidence intervals (5000 resamples) (Table A4) were computed for indirect effects.

The hypothesized path model was tested using Mplus 8.4 (Muthén & Muthén, 2017) with maximum likelihood estimation with robust standard errors (MLR) to account for non-normal data distributions. The analysis examined both direct effects (gender and SES → mathematics achievement) and indirect effects mediated by teacher support, growth mindset, and mathematics anxiety. Model fit was evaluated using multiple indices: (a) comparative fit index (CFI) ≥ 0.95, indicating excellent fit (Hu & Bentler, 1999), (b) root mean square error of approximation (RMSEA) < 0.06, (c) standardized root mean square residual (SRMR) ≤ 0.08.

4. Results

This chapter mainly answers three questions of the research by analyzing the data. Table 1 shows the model-fit indices of the path analysis and all of them meet the requirements.

Fig. 3 presents the direct and indirect relationships existing. The following text will analyze it based on it and the descriptive statistics of student individual characteristics, student learning process and mathematics achievements in the UK (Table A2) in the Appendix A.

4.1. Direct effects of gender and SES on mathematics achievement and psychological factors (RQ1)

Based on the analysis of standardized direct effects (Table A3), this section will address RQ1: How do gender and socioeconomic status directly influence mathematics achievement? Are there gender and SES differences in teacher support, growth mindset, and anxiety?

The analysis conducted on the PISA UK data for 2022 (Table A3) indicates that socioeconomic status (SES) had a significant positive direct effect on mathematics achievement (β=0.30, $p < 0.05$), indicating students from higher SES backgrounds performed better in mathematics. In contrast, gender showed no significant direct effect on achievement (β= -0.00, $p > 0.05 = 0.93$), suggesting comparable performance between boys and girls in the UK.

Further analysis demonstrated that girls reported significantly higher levels of mathematics anxiety than boys (β=-0.24, $p < 0.05$), while SES is inversely proportional to math anxiety (β= -0.08, $p < 0.05$), showing that students with a higher level of SES have lower levels of math anxiety. Regarding growth mindset, boys exhibited slightly stronger growth mindset than girls (β=0.06, $p < 0.05$), and higher SES predicted stronger growth mindset (β=0.07, $p < 0.05$). It is worth noting that teachers provided marginally more support to boys than girls (β=0.04, $p < 0.05$), but SES showed no significant effect on teacher support (β=0.02, $p > 0.05$).

4.2. Mediating mechanisms from gender and SES to mathematics achievement (RQ2)

4.2.1. Relationships among teacher support, growth mindset, anxiety, and achievement

Based on the analysis of standardized direct effects (Table A3), this section will address RQ2: (1)What are the relationships between teacher support, growth mindset, mathematics anxiety and achievement?

First, mathematics teacher support is positively proportional to growth mindset strength (β=0.10, $p < 0.05$), indicating that the more support students receive from their math teachers, the stronger their growth mindset will be. Second, mathematics teacher support is inversely proportional to math anxiety levels (β=-0.16, $p < 0.05$), showing that the more support the math teacher provides, the lower the students' math anxiety will be. Third, growth mindset strength is positively proportional to math achievement (β=0.12, $p < 0.05$), confirming that growth mindset is beneficial for students to get better academic achievement in mathematics. Forth, growth mindset is inversely proportional to math anxiety (β=-0.13, $p < 0.05$), revealing that the more growth mindset students have, the less math anxiety they will experience. Fifth, math anxiety is strongly inversely proportional to mathematics achievement (β = -0.24, $p < 0.05$), indicating that higher levels of math anxiety are associated with significantly lower academic performance in mathematics - this represents the most impactful negative relationship observed in the study. Finally, mathematics teacher support

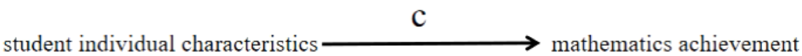

Fig. 1. The total effect of student individual characteristics on mathematics achievement.

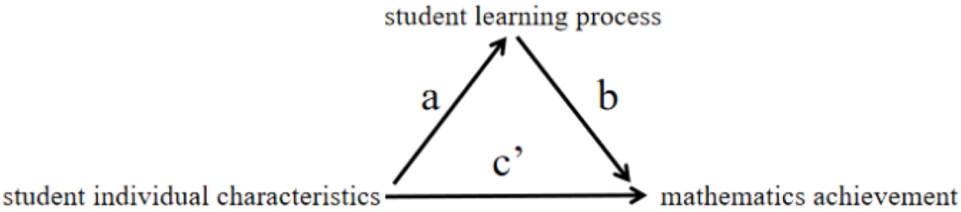

Fig. 2. Hypothesis diagram of mediational analyses among student individual characteristics, student learning process, and mathematics achievement.

Table 1
Model fit test on gender, ESCS, math teacher support, growth mindset, math anxiety, and math achievement.

CFI	RMSEA	SRMR	χ2	df
1.00	0.00	0.00	0.00	0.00

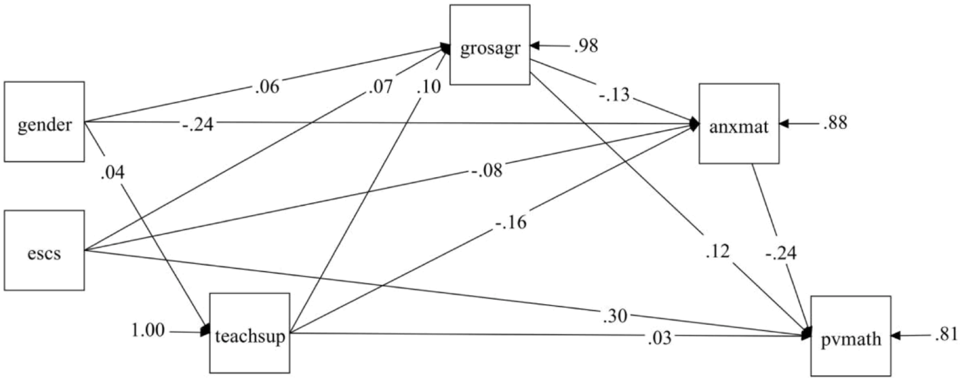

Fig. 3. The mediator model diagram on gender, ESCS, math teacher support, growth mindset, math anxiety, and math achievement.
Note. gender: gender; escs: socioeconomic status; teachsup: mathematics teacher support; grosagr: growth mindset; anxmat: mathematics anxiety; pvmath: mathematics achievement.

shows weak direct proportionality to achievement (β=0.03, $p < 0.05$), showing that the more support students receive from their math teachers, the greater their math achievement will be.

4.2.2. The role of mathematics teacher support in gender/SES-achievement pathways

Based on the indirect path analysis (Table A4), this section will address RQ2: (2)To what extent does mathematics teacher support contribute to the mechanisms linking gender, socioeconomic status, and mathematics achievement, particularly through its influence on growth mindset and mathematics anxiety?

The indirect path analysis (Table A4) reveals three distinct mediating pathways through which mathematics teacher support enhances students' mathematics achievement in the way of gender. The first pathway represents a cognitive-affective dual mediation chain (Gender → Teacher Support → Growth Mindset [cognitive] → Anxiety Reduction [affective] → Achievement, β=0.000, $p < 0.05$), demonstrating that teacher support initially strengthens growth mindset through the cognitive pathway, then improves achievement indirectly by reducing anxiety through the affective pathway. The second pathway reveals a significant cognitive mediation effect (Gender → Teacher Support → Growth Mindset → Achievement, β=0.000, $p < 0.05$), suggesting that the greater support boys receive from math teachers enhances their growth mindset, ultimately improving their math performance. The third pathway demonstrates pure affective mediation (Gender → Teacher Support → Anxiety Reduction → Achievement, β=0.002, $p < 0.05$), indicating that boys receive more support from math teachers, which significantly alleviates their math anxiety and consequently enhances their academic performance. All pathways operate through indirect effects, with the innovative aspect of the first pathway lying in its simultaneous incorporation of both cognitive (growth mindset) and affective (anxiety) mediating mechanisms, comprehensively illustrating how teacher support influences academic performance through different psychological dimensions.

In contrast to gender effects, mathematics teacher support does not mediate the relationship between SES and mathematics achievement (β=0.000, $p > 0.05$). However, a significant mediating pathway emerges from SES to achievement (SES → Growth Mindset → Anxiety Reduction → Achievement, β=0.002, $p < 0.05$). This suggests that students with higher socioeconomic status develop stronger growth mindsets, experience reduced math anxiety, and consequently achieve better math achievement. This suggests that SES-linked advantages operate independently of teacher support, with family background conferring inherent psychological benefits that promote academic success.

5. Discussion

5.1. The direct influence of gender and socioeconomic status on mathematics anxiety and achievement

In this study, gender and socioeconomic status (SES) directly influence math anxiety and achievement. Among them, there is a negative correlation between gender and math anxiety, which means that the math anxiety of boys is generally lower than that of girls. Furthermore, the results indicate that gender did not exhibit a significant direct effect on mathematics achievement. Xie et al. (2019) point out that young women have a higher level of math anxiety than young men, and no gender difference is found in math performance. Therefore, the findings in this study are consistent with the current literature. Some researchers have also used experiments to explain gender differences in math anxiety. Previous research (Xie et al., 2019) suggests that girls tend to experience higher math anxiety than boys, possibly due to societal stereotypes that associate mathematics with male dominance (Dersch et al., 2022). This anxiety, as predicted by Control-Value Theory (Pekrun, 2006), reduces girls' perceived control in math tasks, leading to poorer performance. The findings support this mechanism, showing that gender differences in math anxiety mediate achievement gaps.

This study also found that students' socioeconomic status has a negative correlation with math anxiety and a positive correlation with math achievement. Guzman et al. (2021) selected 451 children with an average age of 5 and a half years in Chile for research and proposed that students' socioeconomic status is inversely correlated with math anxiety. This pattern aligns with existing evidence that children from higher-SES families tend to have greater exposure to early math activities (e.g., number games, spatial toys) during preschool years (Gunderson et al., 2018; James-Brabham, 2022). Such early exposure fosters foundational math skills and positive attitudes, which may reduce later math anxiety by enhancing perceived competence (Pellizzoni et al., 2022). While Chile exhibits higher socioeconomic inequality (Gini coefficient = 0.46 vs. UK's 0.34; World Bank, 2023), the UK findings align with Guzmán et al.'s (2021) Chilean study in suggesting that SES influences math anxiety through early math exposure. This implies that despite contextual differences, resource disparities similarly shape children's foundational skills and emotional responses to math. In addition, James-Brabham (2022) did not explicitly show that there was a positive relationship between socioeconomic status and math achievement but did find that parents' beliefs about the importance of mathematics affect the frequency of family math activities with their children and that families of higher socioeconomic status place more emphasis on math learning than families of lower socioeconomic status, and therefore engage in more family math activities. Therefore, it is possible to explain the positive correlation between socioeconomic status and math achievement from the aspects of socioeconomic status, the degree of emphasis on math education, and the frequency of family math activities.

5.2. The mediating role of student learning process in the influence of gender and socioeconomic status on mathematics achievement

This study found that in student learning process (including mathematics teacher support, growth mindset, and mathematics anxiety), growth mindset and mathematics anxiety play a mediating role in gender, socioeconomic status, and math achievement. Mathematics teacher support did not play a mediating role in the effect of gender and socioeconomic status on mathematics achievement.

On the one hand, although there is a positive correlation between mathematics teacher support and gender, it is not used as one of the mediators of gender and socioeconomic status affecting math achievement. This suggests that in the UK mathematics teachers give more support to male students than female students, but this does not affect the relationship between gender and maths achievement. Similar to the results of this study, the help and support provided by math teachers to students in the past 20 years were significantly biased toward boys, thus reinforcing the self-concept that girls have low math ability (Lindner et al., 2022). In this way, the gender stereotype of math teachers that girls are not good at learning math may be one of the reasons for giving more support to boys in teaching. On the other hand, math teacher support is not related to students' socioeconomic status, nor does it play a mediating role in the influence of socioeconomic status on math achievement. This means that teachers do not discriminate in providing support because students come from families of different socioeconomic statuses. In contrast, a study showed that math teachers have negative attitudes toward male students from lower socioeconomic status and may provide less effective help (Auwarter & Aruguete, 2008). It can be seen that the support of mathematics teachers will change with the development of society. Nearly two decades of social development, so that once biased mathematics teachers to more in line with the concept of educational equity.

Growth mindset acts as a mediator in the influence of both gender and socioeconomic status on mathematics achievement. According to the results of this study, boys have more growth mindsets than girls, and having more growth mindsets can lead to better mathematics achievement. Gender differences in growth mindset align with prior research. Boys are more likely to associate effort with ability improvement (Dweck, 2006), possibly because they receive more explicit feedback linking effort to success in STEM fields (Gunderson et al., 2018). Conversely, girls may internalize fixed-ability stereotypes earlier, as shown in experimental studies where growth mindset interventions significantly reduced gender gaps in math achievement (Degol et al., 2018; Yeager & Dweck, 2020). The results support these patterns, with boys reporting stronger growth mindset despite comparable math performance. Furthermore, growth mindset can enhance students' confidence and perseverance, thereby increasing their likelihood of academic success. At present, no other researchers have conducted research on the relationship between growth mindset, gender, and math achievement. So this discovery fills a gap. Through the discovery of this mechanism, in mathematics teaching, students, especially girls, can be appropriately trained with growth mindset, which is expected to improve students' mathematics achievement. Among the findings, students from families with higher socioeconomic status have more growth mindset, which is associated with better mathematics achievement. Students from higher socioeconomic backgrounds may develop a stronger growth mindset due to early exposure to enriched mathematical learning environments and family support. This mindset, in turn, fosters confidence and diligence, enhancing their academic success. Similar to this research result, it was proved that growth mindset is an important factor in explaining the relationship between socioeconomic status and academic achievement (Destin et al., 2019). Therefore, the cultivation of growth mindset for students, especially students in lower socioeconomic status, is conducive to improving their mathematics achievement.

Mathematics anxiety also acts as a mediator in the influence of both gender and socioeconomic status on mathematics achievement. In this study, boys have less mathematics anxiety than girls, and less math anxiety is associated with better math performance. This conclusion is consistent with the results of many current studies. Math anxiety plays a mediating role between gender and math performance through the analysis of PISA 2012 data, and math anxiety is more harmful to boys than girls (Cox & Jacobson, 2020). It is likely that girls have more math anxiety because of family and social stereotypes about female math. According to the current British research data, easing the mathematics anxiety of students, especially girls, is conducive to improving students' mathematics achievement. The results show that students with higher socioeconomic status have lower math anxiety and lower math anxiety is associated with better math scores. This finding can be attributed to the fact that students from higher socioeconomic backgrounds experience lower anxiety levels, as they benefit from greater life satisfaction, material stability, and early exposure to quality preschool math education. At present, there are no detailed studies on the relationship between socioeconomic status, mathematics anxiety, and achievement. The results of this study fill the gap. Mathematics anxiety plays a mediating role in socioeconomic status and mathematics achievement. Therefore, easing the math anxiety of students, especially those of lower socioeconomic status, can improve their mathematics achievement.

5.3. Teacher support as a mediator between gender, SES, and mathematics achievement: indirect influences via growth mindset and mathematics anxiety

Teacher support has always been an integral part of the teaching process. Two mediator systems for teacher support previously proposed in this study, gender to growth mindset to mathematics anxiety to mathematics achievement, socioeconomic status to teacher support to growth mindset to mathematics anxiety to mathematics achievement, through the research of the PISA 2022 database, is concluded that the former exists and the latter does not exist. This means that teacher support only plays a mediator role in the indirect effect of gender on math achievement through growth mindset and math anxiety, but not in the indirect effect of socioeconomic status on math achievement through growth mindset and math anxiety.

First, this study found a direct relationship between gender and teacher support, but not between students' socioeconomic status and teacher support. In the UK, teachers offer more maths support to boys than to girls but do not offer different support depending on their socioeconomic status. The tendency of mathematics teachers to provide greater support to male students may reflect pervasive gender stereotypes within educational environments. But gender differences in math achievement are reinforced by gender role differences, male dominance, sexual harassment, religious marriage teachings, social stereotypes, and parental preference for boys (Mapulanga & Bwalya, 2025). In this way, mathematics teachers may be influenced by these factors to believe that boys are generally better at mathematics than girls, or that boys have greater mathematical potential than girls, and are therefore willing to provide more help to boys, which has led to the current phenomenon in the UK. The absence of a significant relationship between math teacher support and students' socioeconomic status can be attributed to several factors. Most mathematics teachers are unable to learn about students' families and their socioeconomic status in the classroom and the contact process with students. Secondly, the idea of educational equity has been pursued in Britain. Not discriminating against students based on their socioeconomic status is the most basic thing for educational equity. This also reflects that British mathematics teachers have a good sense of educational equity and high moral quality.

The mediator mechanism "gender to teacher support to growth mindset to mathematics anxiety to mathematics achievement" found in this study contributes to the effect of mathematics teaching and the improvement of students' mathematical ability. A simple explanation of this mechanism is that math teachers will provide more effective support and help to boys, resulting in more growth mindset for boys than girls and more growth mindset can alleviate students' mathematics anxiety, and less mathematics anxiety is more conducive to boys' better math achievement. Therefore, in the process of mathematics teaching, teachers should provide more help to students to improve their growth mindset and reduce math anxiety, thus improving mathematics achievement. According to the item of mathematics teacher support mentioned in the PISA 2022 database, mathematics teacher support mainly refers to sufficient interest in students' learning, and timely provision of help when students need it and until students understand it. Teacher support can indirectly improve students' math performance by doing the above points. But the findings also have advice for maths teachers in the UK. In education and teaching, gender equity has always been an important issue, which is conducive to maintaining educational equity and social equity. Maths teachers should therefore focus on ensuring fairness in their future help, rather than providing more robust support to boys.

6. Limitations and further research directions

Some limitations of this study need to be addressed in future studies. First of all, although PISA is a large-scale international education evaluation project, its sample selection and data collection methods may not fully represent the whole student group in the UK, especially some marginalized or minority groups. Therefore, in future relevant studies, the sample size should be expanded as much as possible, including recruiting more volunteers to participate and improving the representativeness of the sample. In addition, although

PISA itself adopts PPS sampling (Probability Proportionate to Size Sampling) method, the representativeness of samples has been considered to a certain extent. However, to better control the heterogeneity of samples, multi-stage sampling or stratified sampling methods are considered. Databases from other sources, such as national education statistics, school records, and market research data, can also be combined to supplement and validate PISA data. Second, all the variables in the study and their relationships may be influenced by different school environments and cultural contexts. For example, the cultural background of Wales and the fact that mathematics is taught in a language other than English in some schools may have influenced the results. Most importantly, the findings require cautious interpretation regarding regional generalizability. The UK's devolved education systems - notably Scotland's competency-based Curriculum for Excellence versus England's standardized testing regime - likely create distinct learning environments that differentially shape teacher support and student anxiety (Jerrim, 2021). This underscores the need for future studies to utilize region-specific datasets like England's National Pupil Database, enabling systematic comparisons across all four UK nations. Therefore, in future studies, the four regions of the UK can be studied and compared in more detail to get clearer results.

7. Conclusion

This is the first empirical study to combine students' gender, socioeconomic status, mathematics teacher support, growth mindset, and mathematics anxiety to indicate their contributions to student mathematics achievement. Through in-depth analysis of the relationship between gender, socioeconomic status, and math achievement, as well as the function mechanism of mathematics teacher support, growth mindset, and mathematics anxiety as mediating variables, this study not only reveals the problems and challenges faced by math teaching in the UK, but also provides a new perspective to understand the differences in students' math achievement.

Mathematics teacher support plays a significant role in the effects of gender on math achievement. By providing targeted guidance and support, teachers are able to help students overcome learning barriers, especially when faced with the challenging subject of mathematics. In addition, the cultivation of a growth mindset has been proven to be an effective strategy for reducing mathematics anxiety. By encouraging students to develop positive self-beliefs and believe in the value of hard work and progress, teachers can help students develop a more resilient learning attitude and thus achieve better results in math learning. At the same time, this study also emphasizes the importance of math anxiety as a mediating variable. High math anxiety level often prevents students from succeeding, and effective intervention measures to reduce the anxiety level can significantly improve students' learning efficiency and performance. In this process, the professional support of mathematics teachers and the cultivation of growth mindset play a crucial role.

Therefore, schools should encourage teachers to adopt more flexible teaching strategies to meet the needs of students from different gender and socioeconomic backgrounds. At the same time, it is important to strengthen the professional development of teachers to improve their understanding and coping ability of growth mindset and math anxiety. In addition, from the perspective of educational equity, all students should enjoy equal educational opportunities and resources. However, according to the current PISA 2022 database, mathematics teachers in the UK will provide more effective support to boys, which will exacerbate the gap between male and female math achievement in the long term. Therefore, teachers need to be aware that gender biases or stereotypes can have a negative impact on students' academic development. Through training and education, teachers can raise awareness of gender equality, learn how to treat each student more equitably, and provide them with appropriate support.

CRediT authorship contribution statement

Mengying Huang: Writing – original draft, Resources, Supervision, Writing – review & editing. **Xin Liu:** Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijer.2025.102666.

Appendix A

Table A1
Item information of student individual characteristics and student learning.

Item	Item Wording	Variable Coding
	Student (Standardized) Gender	
ST004D01T	Student (Standardized) Gender	0 Female 1 Male
	Index of Economic, Social and Cultural Status (ESCS)	
	Home Possessions	
ST250Q01JA	Which of the following are in your [home]: A room of your own.	1 Yes
ST250Q02JA	Which of the following are in your [home]: A computer (laptop, desktop, or tablet) that you can use for school work.	2 No

(continued on next page)

Table A1 (*continued*)

Item	Item Wording	Variable Coding
ST250Q03JA	Which of the following are in your [home]: Educational Software or Apps.	
ST250Q04JA	Which of the following are in your [home]: Your own [cell phone] with Internet access (e.g. smartphone).	
ST250Q05JA	Which of the following are in your [home]: Internet access (e.g. Wi-fi) (excluding through smartphones).	
ST250D06JA	Which of the following are in your home? United Kingdom (Excl. Scotland):A games console such as a PlayStation 4 or Nintendo Switch.	
ST250D07JA	Which of the following are in your home? United Kingdom (Excl. Scotland):A smart speaker such as Amazon Echo or Google Home.	
ST251Q01JA	How many of these items are there at your [home]: Cars, vans, or trucks.	1 None
ST251Q02JA	How many of these items are there at your [home]: Mopeds or motorcycles.	2 One
ST251Q03JA	How many of these items are there at your [home]: Rooms with a bath or shower.	3 Two
ST251Q04JA	How many of these items are there at your [home]: Rooms with a [flush toilet].	4 Three or more
ST251Q06JA	How many of these items are there at your [home]: Musical instruments (e.g. guitar, piano, [country-specific example]).	
ST251Q07JA	How many of these items are there at your [home]: Works of art (e.g. paintings, sculptures, [country-specific example]).	
ST253Q01JA	How many [digital devices] with screens are there in your [home]?	1 There are no ⟨digital devices⟩ with screens. 2 One 3 Two 4 Three 5 Four 6 Five 7 6 to 10 8 >10
ST254Q01JA	How many of the following [digital devices] are in your [home]: Televisions.	1 None
ST254Q02JA	How many of the following [digital devices] are in your [home]: Desktop computers.	2 1 or 2
ST254Q03JA	How many of the following [digital devices] are in your [home]: Laptop computers or notebooks.	3 3 - 5
ST254Q04JA	How many of the following [digital devices] are in your [home]: Tablets (e.g. [iPad®], [BlackBerry® Playbook™]).	4 >5 5 I don't know.
ST254Q05JA	How many of the following [digital devices] are in your [home]: E-book readers (e.g. [Kindle™], [Kobo], [Bookeen]).	
ST254Q06JA	How many of the following [digital devices] are in your [home]: [Cell phones] with Internet access (i.e. smartphones).	
ST255Q01JA	How many books are there in your [home]?	1 There are no books. 2 1 - 10 books 3 11 - 25 books 4 26 - 100 books 5 101 - 200 books 6 201 - 500 books 7 >500 books
ST256Q01JA	How many of these books at [home]: Religious books (e.g. [Bible], [Example 2]).	1 None
ST256Q02JA	How many of these books at [home]: Classical literature (e.g. [Shakespeare], [Example 2]),	2 1 or 2
ST256Q03JA	How many of these books at [home]: Contemporary literature.	3 3 - 5
ST256Q06JA	How many of these books at [home]: Books on science.	4 >5
ST256Q07JA	How many of these books at [home]: Books on art, music, or design.	5 I don't know.
ST256Q08JA	How many of these books at [home]: [Technical reference books].	
ST256Q09JA	How many of these books at [home]: Dictionaries.	
	Index of Economic, Social and Cultural Status (ESCS)	
	Highest Education of Parents in Years	
ST005Q01JA	What is the [highest level of schooling] completed by your mother?	1 ⟨ISCED level 3.4⟩ 2 ⟨ISCED level 3.3⟩ 3 ⟨ISCED level 2⟩ 4 ⟨ISCED level 1⟩ 5 She did not complete ⟨ISCED level 1⟩.
ST006Q01JA	Does your mother have any of the following qualifications: [ISCED level 8].	1 Yes
ST006Q02JA	Does your mother have any of the following qualifications: [ISCED level 7].	2 No
ST006Q03JA	Does your mother have any of the following qualifications: [ISCED level 6].	
ST006Q04JA	Does your mother have any of the following qualifications: [ISCED level 5].	
ST006Q05JA	Does your mother have any of the following qualifications: [ISCED level 4].	
ST007Q01JA	What is the [highest level of schooling] completed by your father?	1 ⟨ISCED level 3.4⟩ 2 ⟨ISCED level 3.3⟩ 3 ⟨ISCED level 2⟩ 4 ⟨ISCED level 1⟩ 5 He did not complete ⟨ISCED level 1⟩.
ST008Q01JA	Does your father have any of the following qualifications: [ISCED level 8].	1 Yes
ST008Q02JA	Does your father have any of the following qualifications: [ISCED level 7].	2 No
ST008Q03JA	Does your father have any of the following qualifications: [ISCED level 6].	

(*continued on next page*)

Table A1 (*continued*)

Item	Item Wording	Variable Coding
ST008Q04JA	Does your father have any of the following qualifications: [ISCED level 5].	
ST008Q05JA	Does your father have any of the following qualifications: [ISCED level 4].	
	Index of Economic, Social and Cultural Status (ESCS)	
	Highest Parental Occupation Status	
OCOD1	ISCO-08 Occupation code - Mother	From 0 to 9999 represents different
OCOD2	ISCO-08 Occupation code - Father	occupations
	Mathematics Teacher Support	
ST270Q01JA	How often: The teacher shows an interest in every student's learning.	1 Every lesson
ST270Q02JA	How often: The teacher gives extra help when students need it.	2 Most lessons
ST270Q03JA	How often: The teacher helps students with their learning.	3 Some lessons
ST270Q04JA	How often: The teacher continues teaching until the students understand.	4 Never or almost never
	Mathematics Anxiety	
ST292Q01JA	Agree/disagree: I often worry that it will be difficult for me in mathematics classes.	1 Strongly agree
ST292Q02JA	Agree/disagree: I get very tense when I have to do mathematics homework.	2 Agree
ST292Q03JA	Agree/disagree: I get very nervous doing mathematics problems.	3 Disagree
ST292Q04JA	Agree/disagree: I feel helpless when doing a mathematics problem.	4 Strongly disagree
ST292Q05JA	Agree/disagree: I worry that I will get poor [marks] in mathematics.	
ST292Q06JA	Agree/disagree: I feel anxious about failing in mathematics.	
	Growth mindset	
ST263Q02JA	Agree/disagree: Your intelligence is something about you that you cannot change very much.	1 Strongly disagree
ST263Q04JA	Agree/disagree: Some people are just not good at mathematics, no matter how hard they study.	2 Disagree
ST263Q06JA	Agree/disagree: Some people are just not good in [test language], no matter how hard they study.	3 Agree
ST263Q08JA	Agree/disagree: Your creativity is something about you that you cannot change very much.	4 Strongly agree

Table A2
PISA2022 descriptive statistics of student individual characteristics, student learning process and mathematics achievements in the UK.

Label	N	Minimum	Maximum	M	SD
Student Standardized Gender	12,972	0	1	.51	.50
Index of economic, social and cultural status (ESCS)	11,084	−5.91	3.50	.12	.89
Mathematics Teacher Support (TEACHSUP)	10,600	−2.91	1.56	.23	1.10
Mathematics Anxiety (ANXMAT)	10,072	−2.39	2.64	.09	1.12
Growth Mindset (GROSAGR)	10,752	−3.60	3.36	.10	.96
Plausible Value 1 in Mathematics (PV1MATH)	12,972	154.86	853.17	482.54	95.09
Plausible Value 2 in Mathematics (PV2MATH)	12,972	147.15	865.50	481.35	95.36
Plausible Value 3 in Mathematics (PV3MATH)	12,972	128.59	832.72	482.04	95.05
Plausible Value 4 in Mathematics (PV4MATH)	12,972	133.77	868.14	481.57	95.77
Plausible Value 5 in Mathematics (PV5MATH)	12,972	124.00	857.15	482.11	95.65
Plausible Value 6 in Mathematics (PV6MATH)	12,972	140.91	856.01	481.82	95.37
Plausible Value 7 in Mathematics (PV7MATH)	12,972	172.18	872.40	481.37	95.82
Plausible Value 8 in Mathematics (PV8MATH)	12,972	119.38	847.11	482.46	95.66
Plausible Value 9 in Mathematics (PV9MATH)	12,972	176.56	842.67	481.36	94.55
Plausible Value 10 in Mathematics (PV10MATH)	12,972	128.30	915.23	481.61	95.57

Table A3
Standardized direct effects of gender, socioeconomic status, mathematics teacher support, growth mindset and mathematics anxiety on mathematics achievement.

Pathway	Estimate	P-Value
GENDER→PVMATH	−0.00	0.93
GROSAGR→PVMATH	0.12**	0.00
ANXMAT→PVMATH	−0.24**	0.00
ESCS→PVMATH	0.30**	0.00
TEACHSUP→PVMATH	0.03**	0.04
GENDER→ANXMAT	−0.24**	0.00
GROSAGR→ANXMAT	−0.13**	0.00
ESCS→ANXMAT	−0.08**	0.00
TEACHSUP→ANXMAT	−0.16**	0.00
GENDER→GROSAGR	0.06**	0.00
ESCS→GROSAGR	0.07**	0.00
TEACHSUP→GROSAGR	0.10**	0.00
GENDER→TEACHSUP	0.04**	0.01
ESCS→TEACHSUP	0.02	0.30

Note. ** indicates $p < 0.05$. The confidence interval is based on the Bootstrap method (with 5000 sampling iterations). Variable abbreviations are explained below.
GENDER: gender; ESCS: socioeconomic status; TEACHSUP: mathematics

teacher support; GROSAGR: growth mindset; ANXMAT: mathematics anxiety; PVMATH: mathematics achievement.

Table A4
Standardized indirect effects of gender, socioeconomic status, mathematics teacher support, growth mindset and mathematics anxiety on mathematics achievement.

Mediation Pathway	Estimate	Boot SE	95 % Bootstrap CI	P-Value
GENDER→GROSAGR→PVMATH	0.007**	0.002	[0.003, 0.011]	0.001
GENDER→ANXMAT→PVMATH	0.056**	0.006	[0.044, 0.068]	0.000
GENDER→TEACHSUP→PVMATH	0.001	0.001	[−0.001, 0.003]	0.091
GENDER→TEACHSUP→GROSAGR→PVMATH	0.000**	0.000	[0.000, 0.001]	0.019
GENDER→GROSAGR→ANXMAT→PVMATH	0.002**	0.001	[0.001, 0.003]	0.001
GENDER→TEACHSUP→ANXMAT→PVMATH	0.002**	0.001	[0.000, 0.004]	0.013
GENDER→TEACHSUP→GROSAGR→ANXMAT→PVMATH	0.000**	0.000	[0.000, 0.001]	0.021
ESCS→GROSAGR→PVMATH	0.008**	0.002	[0.004, 0.012]	0.000
ESCS→ANXMAT→PVMATH	0.018**	0.004	[0.010, 0.026]	0.000
ESCS→TEACHSUP→PVMATH	0.001	0.001	[−0.001, 0.001]	0.373
ESCS→TEACHSUP→GROSAGR→PVMATH	0.000	0.000	[−0.001, 0.001]	0.308
ESCS→GROSAGR→ANXMAT→PVMATH	0.002**	0.001	[0.001, 0.003]	0.000
ESCS→TEACHSUP→ANXMAT→PVMATH	0.001	0.001	[−0.001, 0.003]	0.295
ESCS→TEACHSUP→GROSAGR→ANXMAT→PVMATH	0.000	0.000	[−0.001, 0.001]	0.305

Note. ** indicates $p < 0.05$. Variable abbreviations are explained below.
GENDER: gender; ESCS: socioeconomic status; TEACHSUP: mathematics teacher support; GROSAGR: growth mindset; ANXMAT: mathematics anxiety; PVMATH: mathematics achievement.

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